



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2006
YEAR 12
ASSESSMENT TASK #2

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.
- Hand in your answer booklets in 3 sections. Section A (Question 1), Section B (Question 2) and Section C (Question 3)

Total Marks – 74

- Attempt questions 1-3
- All sections are not of equal value

Examiner: *R.Boros*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Section A – Start a new booklet**Marks****Question 1. (22 marks)****a)**

Differentiate the following expressing your answer in its simplest form:

(i) $y = \ln(x+1)$ 2

(ii) $y = \sin^{-1} 2x$ 2

b) Evaluate (leaving your answer in exact form).

(i) $\int_0^2 \frac{t}{t^2+1} dt$ 2

(ii) $\int_0^1 \frac{dx}{\sqrt{2-x^2}}$ 2

c) Find the equation (written in general form) of the normal to the curve 3

$y = \tan^{-1} 2x$, at the point where $x = \frac{1}{2}$.

d) Find the equation (written in general form) of the tangent to $y = e^{\tan^{-1} x}$ at the point where the curve cuts the y axis. 3**e)** (i) Differentiate $x \tan^{-1} 3x$ with respect to x . 4

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{1}{3}} \left(\tan^{-1} 3x + \frac{3x}{1+9x^2} \right) dx$

f) The function $y = e^{-kx}$ satisfies $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 3y = 0$ 4

(i) Show that $k^2 - 4k + 3 = 0$

(ii) Hence, find the possible values of k .**End of Section A**

Section B – Start a new booklet**Question 2. (25 marks)**

- a) Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$. 1
- b) Evaluate $\cos\left(\sin^{-1}\left(-\frac{1}{2}\right)\right)$, leaving your answer in exact form. 1
- c) Without using a calculator, show that: 3
- $$\tan^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(\frac{1}{4}\right) = \tan^{-1}\left(\frac{2}{9}\right)$$
- d) Solve the equation: 3
- $$\ln(x^3 + 19) = 3\ln(x + 1).$$
- e) Taking $x = 3$ as the first approximation to the root of $x^2 - \ln x - 10 = 0$, use Newtons' Method once to find another approximation correct to two decimal places. 3
- f) What is the domain of the function $f(x) = \frac{x}{\ln(x-1)}$. 2
- g) Consider the function $y = 2\sin^{-1}\left(\frac{x}{3}\right)$. 5
- (i) State the domain.
- (ii) State the range.
- (iii) Hence, or otherwise, sketch the function.

Question 2 continues overleaf.

- h)** A function is given by the rule $f(x) = \frac{x+1}{x+2}$. 2

Find the rule for the inverse function $f^{-1}(x)$.

- i)** The diagram below shows the graph of the function $y = xe^{-x}$. 5

A is a stationary point on the curve.

(i) Show that A is the point $\left(1, \frac{1}{e}\right)$.

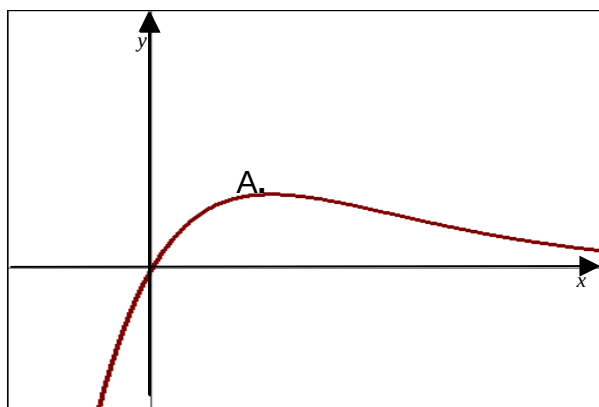
(ii) State the range of the function $y = xe^{-x}$

(iii) How many real roots are there to the equation $xe^{-x} = k$ if

(a.) $0 < k < \frac{1}{e}$

(b.) $k \leq 0$

(c.) $k > \frac{1}{e}$

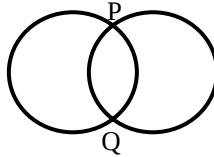


End of Section B

Section C – Start a new booklet**Question 3. (27 marks)**

- a) A cosine curve has an amplitude of 5 and a period of 3π . It has a minimum turning point at $(0, 5)$. Find its equation. 3

- b) 6



Two circles each with radius 2cm intersect at P and Q. The common chord PQ subtends an angle θ radians at each centre.

- (i) Show that the area $A \text{ cm}^2$ of the overlapping part of the circles is given by $A = 4\theta - 4\sin \theta$

- (ii) If the three regions shown in the diagram all have the same area, show

$$\text{that } \theta - \sin \theta - \frac{\pi}{2} = 0$$

- c) The area between the curve $y = \sin^2 x$ and the x -axis between $x = 0$ and $x = \frac{\pi}{2}$, is rotated through one complete revolution about the x -axis. 5

- (i) Find the exact value of the **area** involved.
- (ii) Use Simpson's Rule with 3 function values to find an approximation to the volume of the solid of revolution leaving your answer in terms of π .

Question 3 continues overleaf.

d) Given $f(x) = \frac{8}{4+x^2}$. 8

- (i) Show that $f(x)$ is an even function.
- (ii) Sketch a graph of $y = f(x)$.
- (iii) The line $y = 1$ meets the curve at 2 points P and Q. Determine the x coordinates of P and Q.
- (iv) Calculate the exact area of the region enclosed by the interval PQ and the arc PQ of the curve.
- (v) The region in (iv) makes a revolution about the y-axis, show that the volume of the solid formed is $4\pi(2\ln 2 - 1) \text{ units}^3$.

e) (i) By using the Principle of Mathematical Induction, prove that: 5

$$6(1^2 + 2^2 + 3^2 + \dots + n^2) = n(n+1)(2n+1)$$

- (ii) Hence, find the value of the limit:

$$\lim_{n \rightarrow \infty} \left(\frac{1^2 + 2^2 + 3^2 + \dots + n^2}{n^3} \right)$$

End of Examination

Question 1.

a) i) $\frac{dy}{dx} = \frac{1}{5x+1}$

ii) $y = \sin^{-1} \frac{x}{\frac{1}{2}}$

$$\frac{dy}{dx} = \frac{1}{\sqrt{(\frac{1}{2})^2 - x^2}}$$

$$= \frac{1}{\sqrt{\frac{1}{4} - \frac{4x^2}{4}}}$$

$$= \frac{2}{\sqrt{1-4x^2}}$$

b) i) $\int_0^2 \frac{dt}{t^2+1} = \frac{1}{2} \int_0^2 \frac{2t}{t^2+1} dt$

$$= \frac{1}{2} \left[\ln(t^2+1) \right]_0^2$$

$$= \frac{1}{2} (\ln 5 - \ln 1)$$

$$= \frac{1}{2} \ln 5$$

ii) $\int_0^1 \frac{dx}{\sqrt{2-x^2}} = \int_0^1 \frac{1}{\sqrt{2-x^2}} dx$

$$= \left[\sin^{-1} \frac{x}{\sqrt{2}} \right]_0^1$$

$$= \sin^{-1} \frac{1}{\sqrt{2}} - \sin^{-1} 0 = \frac{\pi}{4}$$

c) Normal to $y = \tan^{-1} 2x$ at $x = \frac{1}{2}$

$$y = \frac{1}{2} (2 \tan^{-1} \frac{x}{\frac{1}{2}})$$

$$\frac{dy}{dx} = \frac{1}{2} \left(\frac{1}{(\frac{1}{2})^2 + x^2} \right)$$

$$= \frac{1}{2} \times \frac{1}{\frac{1}{4} + \frac{4x^2}{4}}$$

$$= \frac{2}{1+4x^2}$$

Gradient at $x = \frac{1}{2}$.

$$m = \frac{2}{1+4(\frac{1}{2})^2}$$

$$= 1$$

$$m_{\perp} = -1.$$

Point on curve at $x = \frac{1}{2}$.

$$y = \tan^{-1}(1).$$

$$= \frac{\pi}{4}$$

$$(\frac{1}{2}, \frac{\pi}{4}).$$

Eqn of normal.

$$y - \frac{\pi}{4} = -(x - \frac{1}{2})$$

$$y - \frac{\pi}{4} = -x + \frac{1}{2}$$

$$x + y - \frac{\pi}{4} - \frac{1}{2} = 0$$

$$4x + 4y - \pi - 2 = 0.$$

Tangent to

d) $y = e^{\tan^{-1}x}$ at $x=0$.

$$\frac{dy}{dx} = e^{\tan^{-1}x} \left(\frac{1}{1+x^2} \right)$$

Value at $x=0$.

$$m = e^{\tan^{-1}0} \left(\frac{1}{1+0^2} \right)$$

$$= e^0$$
$$= 1.$$

Point on curve at $x=0$

$$y = e^{\tan^{-1}0}$$

$$= 1$$
$$(0, 1).$$

Eqn of tangent.

$$y - 1 = x$$

$$x - y + 1 = 0.$$

e) i) $f(x) = x \tan^{-1} 3x$

$$= \frac{1}{3} x \times 3 \tan^{-1} \frac{2x}{\frac{1}{3}}$$

$$f'(x) = \frac{1}{3} \times 3 \tan^{-1} \frac{2x}{\frac{1}{3}} + \frac{1}{3} x \frac{1}{(\frac{1}{3})^2 + x^2}$$

$$= \tan^{-1} 3x + \frac{1}{3} \times \frac{9x}{1+9x^2}$$

$$= \tan^{-1} 3x + \frac{3x}{1+9x^2}$$

ii) $\int_0^1 \tan^{-1} 3x + \frac{3x}{1+9x^2} dx = \left[x \tan^{-1} 3x \right]_0^1$

$$= \frac{1}{3} \tan^{-1} 1 - 0$$

$$= \frac{\pi}{12}$$

f) i) $y = e^{-kx}$

$$\frac{dy}{dx} = -ke^{-kx}$$

$$\frac{d^2 y}{dx^2} = k^2 e^{-kx}$$

So $k^2 e^{-kx} - 4ke^{-kx} + 3e^{-kx} = 0$

Since $e^{-kx} \neq 0$ $\therefore$ we can divide through by e^{-kx}

$$k^2 - 4k + 3 = 0$$

f) ii) $k^2 - 4k + 3 = 0$

$$(k-3)(k-1) = 0$$

$$k = 3, 1.$$

QUESTION 2

(a) $\lim_{x \rightarrow 0} \frac{\sin 2x}{x} = 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$

$$= 2$$

(b) $\cos \left[\sin^{-1} \left(-\frac{1}{2} \right) \right] = \cos \left(-\frac{\pi}{6} \right)$
 $= \sqrt{3}/2$

(c) Let $x = \tan^{-1}(\frac{1}{2})$ and $y = \tan^{-1}(\frac{1}{4})$
 $\tan x = \frac{1}{2}$ $\tan y = \frac{1}{4}$
 $\tan(x-y) = \frac{\frac{1}{2} - \frac{1}{4}}{1 + \frac{1}{2} \cdot \frac{1}{4}}$

$$= \frac{2}{9}$$

$$x - y = \tan^{-1} \left(\frac{2}{9} \right)$$

$$\tan^{-1} \left(\frac{1}{2} \right) - \tan^{-1} \left(\frac{1}{4} \right) = \tan^{-1} \left(\frac{2}{9} \right)$$

(d) $\ln(x^3 + 19) = 3 \ln(x+1)$

$$x^3 + 19 = (x+1)^3$$

$$x^3 + 19 = x^3 + 3x^2 + 3x + 1$$

$$3x^2 + 3x - 18 = 0$$

$$x^2 + x - 6 = 0$$

$$(x+3)(x-2) = 0$$

$$x = 2 \quad (x > -1)$$

e $f(x) = x^2 \ln x - 10$

$$f'(x) = 2x - \frac{1}{x}$$

$$a_1 = 3 - \frac{3^2 \ln 3 - 10}{6 - \frac{1}{3}}$$

$$\text{approx } x \approx 3.37$$

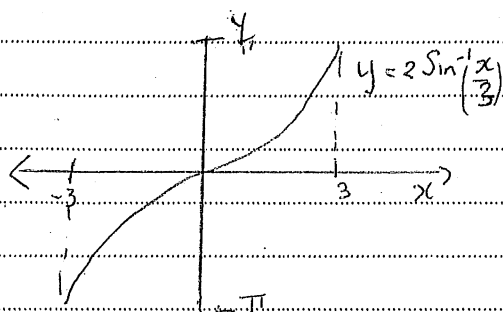
(f) $x-1 > 0 \quad x > 1$

$$\ln(x-1) \neq 0 \quad x-1 \neq 1$$

$$x > 1 + x \neq 2$$

(g) Domain $-3 \leq x \leq 3$

$$\text{Range } -\pi \leq y \leq \pi$$



(h) $f(x) = \frac{x+1}{x+2} = y$

$$\text{Inverse } x = \frac{y+1}{y+2}$$

$$xy + 2x = y + 1$$

$$y(x-1) = 1-2x$$

$$f^{-1}(x) = y = \frac{1-2x}{x-1}$$

(i) $y = xe^{-x}$

(i) $y' = e^{-x} - x \cdot e^{-x}$
 $= e^{-x}(1-x)$

$$y' = 0 \quad x = 1 \quad y = \frac{1}{e}$$

only one solution

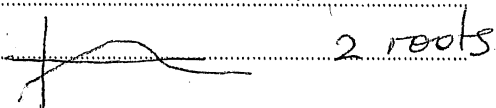
It is the point $(1, \frac{1}{e})$

(ii) range $y \leq \frac{1}{e}$

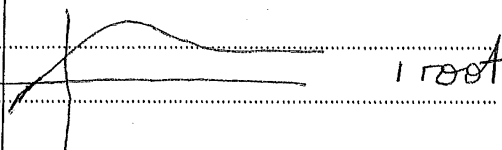
(iii) $xe^{-x} = k$

$$xe^{-x} - k = 0$$

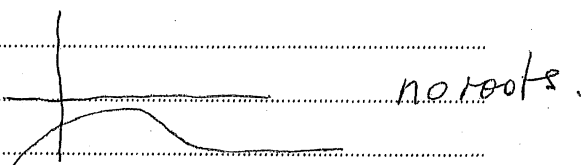
(i) curve moves down less than $\frac{1}{e}$ units



(ii) $xe^{-x} - (-k)$ curve moves up



(iii) Curve moves down more than $\frac{1}{e}$

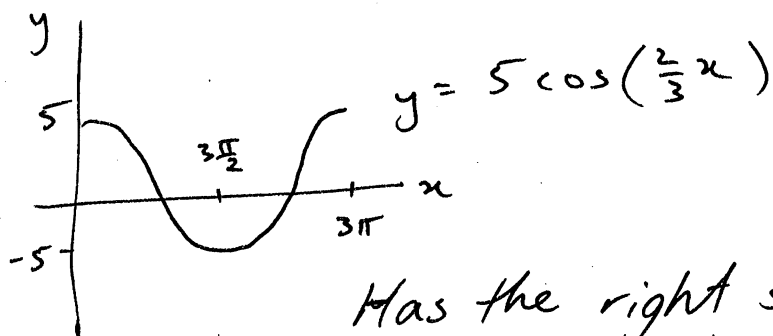


Question 3

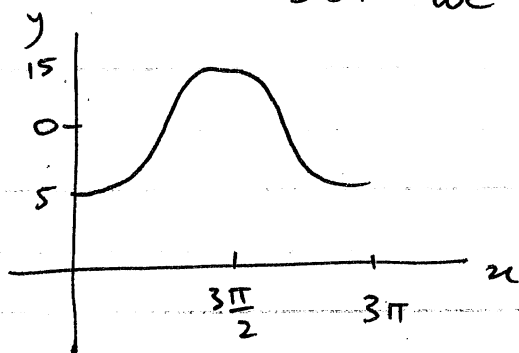
(a) period = $3\pi = \frac{2\pi}{b}$

$$\therefore b = \frac{2}{3}$$

$$a = 5$$



Has the right shape
BUT we want.



The curve we want shifts $y = 5 \cos\left(\frac{2}{3}x\right)$
horizontally $\frac{3\pi}{2}$ units (to left or right)
vertically 10 units (up)

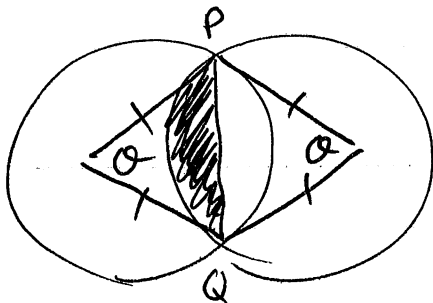
$$\therefore y = 5 \cos\left(\frac{2}{3}\left(x + \frac{3\pi}{2}\right)\right) + 10 \quad \text{OR} \quad y = 5 \cos\left(\frac{2}{3}\left(x - \frac{3\pi}{2}\right)\right) + 10$$

$$y = 5 \cos\left(\frac{2}{3}x + \pi\right) + 10$$

$$y = 5 \cos\left(\frac{2}{3}x - \pi\right) + 10$$

$$\text{OR} \quad y = -5 \cos\left(\frac{2}{3}x\right) + 10$$

(b)



$$\begin{aligned} \text{(i) Area of minor segment (shaded)} &= \frac{1}{2} r^2 (\theta - \sin \theta) \\ &= \frac{1}{2} (2)^2 (\theta - \sin \theta) \\ &= 2 (\theta - \sin \theta) \end{aligned}$$

$$\begin{aligned} \text{Area concerned } A &= 2 (2 (\theta - \sin \theta)) \\ A &= 4 (\theta - \sin \theta) \\ A &= 4\theta - 4\sin \theta \end{aligned}$$

$$\begin{aligned} \text{(ii) Area of circle} &= \pi r^2 \\ &= \pi (2)^2 \\ &= 4\pi \end{aligned}$$

If the three regions have the same area

$$\text{Area of circle} = 2A$$

$$4\pi = 2 (4\theta - 4\sin \theta)$$

$$4\pi = 8\theta - 8\sin \theta$$

$$8\theta - 8\sin \theta - 4\pi = 0$$

$$\theta - \sin \theta - \frac{\pi}{2} = 0$$

$$\begin{aligned}
 (c)(i) \quad A &= \int_0^{\frac{\pi}{2}} \sin^2 x \, dx \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2x) \, dx \\
 &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{2} \left[\frac{\pi}{2} - \cancel{\frac{1}{2} \sin \pi} - \left(0 - \cancel{\frac{1}{2} \sin 0} \right) \right] \\
 &= \frac{\pi}{4} \text{ units}^2
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad V &= \pi \int_0^{\frac{\pi}{2}} y^2 \, dx \\
 &= \pi \int_0^{\frac{\pi}{2}} \sin^4 x \, dx
 \end{aligned}$$

$$f(x) = \sin^4 x$$

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$f(x)$	0	$\frac{1}{4}$	1

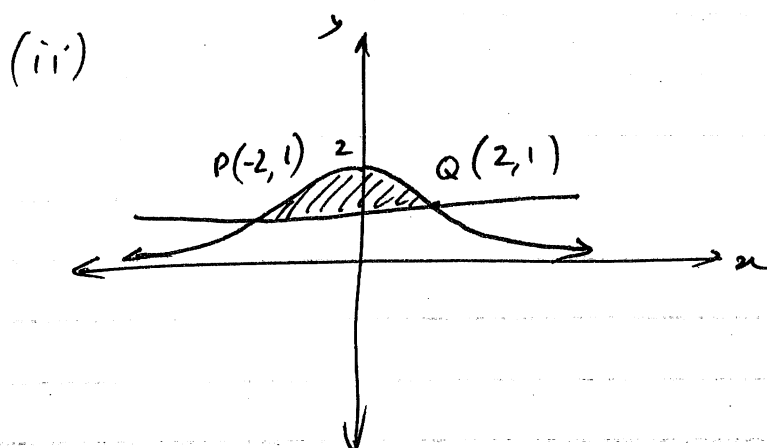
$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \sin^4 x \, dx &\approx \frac{\frac{\pi}{2}}{6} \left[0 + 4\left(\frac{1}{4}\right) + 1 \right] \\
 &= \frac{\pi}{12} [2] \\
 &= \frac{\pi}{6}
 \end{aligned}$$

$$\begin{aligned}
 \therefore V &= \pi \int_0^{\frac{\pi}{2}} \sin^4 x \, dx \\
 &\approx \frac{\pi^2}{6} \text{ units}^3
 \end{aligned}$$

$$(d)(i) \quad f(x) = \frac{8}{4+x^2}$$

$$\begin{aligned} f(-x) &= \frac{8}{4+(-x)^2} \\ &= \frac{8}{4+x^2} \end{aligned}$$

Since $f(x) = f(-x)$
 $f(x)$ is even



(iii) P has x coordinate -2
 Q has x coordinate 2

$$\begin{aligned} (iv) \text{ Area} &= 2 \left[\int_0^2 \frac{8}{4+x^2} dx - 2 \times 1 \right] \\ &= 2 \left[8 \left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^2 - 2 \right] \\ &= 2 \left[4 \tan^{-1} 1 - 4 \cancel{\tan^{-1} 0} - 2 \right] \\ &= 2 \left[4 \cdot \frac{\pi}{4} - 2 \right] \\ &= 2 [\pi - 2] \text{ units}^2 \end{aligned}$$

$$(v) \quad y = \frac{8}{4+x^2}$$

$$4+x^2 = \frac{8}{y}$$

$$x^2 = \frac{8}{y} - 4$$

$$V = \pi \int_1^2 x^2 dy$$

$$V = \pi \int_1^2 \left(\frac{8}{y} - 4 \right) dy$$

$$= \pi \left[8 \ln y - 4y \right]_1^2$$

$$= \pi \left[8 \ln 2 - 4(2) - (8 \ln 1 - 4(1)) \right]$$

$$= \pi [8 \ln 2 - 4]$$

$$= 4\pi (2 \ln 2 - 1)$$

$$(e) \quad 6(1^2 + 2^2 + 3^2 + \dots + n^2) = n(n+1)(2n+1)$$

Prove true for $n=1$

$$\begin{aligned} \text{LHS} &= 6(1)^2 \\ &= 6 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= (1)(1+1)(2(1)+1) \\ &= 1 \cdot 2 \cdot 3 \\ &= 6 \end{aligned}$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ true for $n=1$

Assume true for $n=k$ where k is a positive integer.

$$6(1^2 + 2^2 + 3^2 + \dots + k^2) = k(k+1)(2k+1)$$

Prove true for $n=k+1$

ie Prove $6(1^2 + 2^2 + \dots + k^2 + (k+1)^2) = (k+1)(k+2)(2(k+1)+1)$
 $= (k+1)(k+2)(2k+3)$

$$\text{LHS} = 6(1^2 + 2^2 + \dots + k^2) + 6(k+1)^2$$

$$= k(k+1)(2k+1) + 6(k+1)^2 \quad \text{using assumption.}$$

$$= (k+1)(2k^2 + k + 6k + 6)$$

$$= (k+1)(2k^2 + 7k + 6)$$

$$= (k+1)(2k+3)(k+2)$$

$$= (k+1)(k+2)(2k+3)$$

$$= \text{RHS}$$

$$\begin{array}{r} \times 12 \\ + 7 \\ \hline 3, 4 \end{array}$$

$\therefore$ true for $n=k+1$

If true for $n=k$ it is true for $n=k+1$. Since true for $n=1$ by the principle of mathematical induction it is true for all positive integers n .

(ii) $\lim_{n \rightarrow \infty} \frac{(1^2 + 2^2 + 3^2 + \dots + n^2)}{n^3}$

$$= \lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6n^3}$$

$$= \lim_{n \rightarrow \infty} \frac{2n^3 + 3n^2 + n}{6n^3}$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{2n^3} + \frac{3n^2}{\cancel{n^3}} + \frac{n}{\cancel{n^3}}}{6\cancel{n^3}} = \frac{1}{3}$$